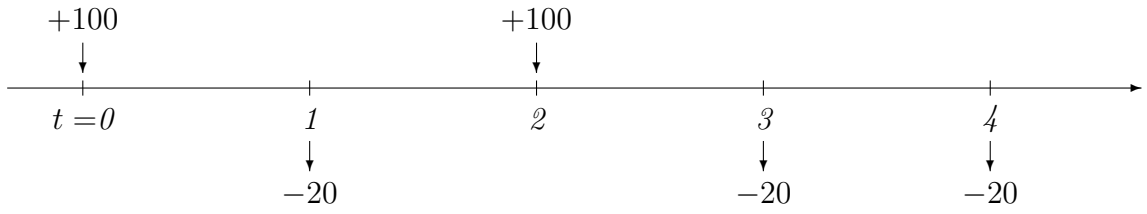
F7AB - Actuarial and Financial Mathematics
Tutorial 2 – Valuing cashflows and annuities

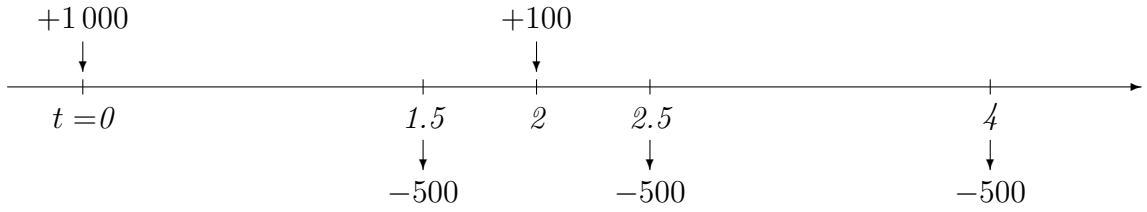
Question 1 The effective rate of interest is 6% per time unit. Cashflows are as follows (the time axis is calibrated in time units):



For these cashflows, calculate:

1. the accumulation at time $t = 5$ time units.
2. the accumulation at time $t = 10$ time units.
3. the present value at time $t = 0$ time units.

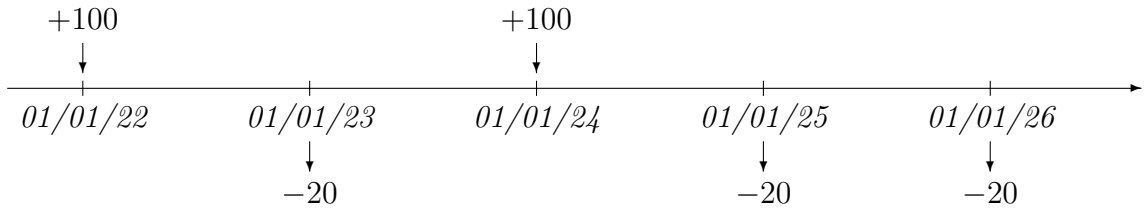
Question 2 The effective rate of interest is 8% per annum. Cashflows are as follows (the time axis is calibrated in years):



For these cashflows, calculate:

1. the total accumulation at time $t = 5$ years.
2. the total present value at time $t = 0$ years.
3. the total value at time $t = 2$ years.

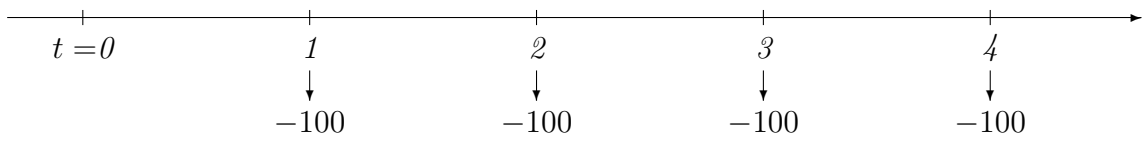
Question 3 The effective rate of interest is 3% per annum during 2022 and 2023, 2% per half-year until 1 July 2025, and 1% per month thereafter. Cashflows are as follows (the time axis is calibrated in calendar time):



For these cashflows, calculate the total:

1. present value on 01/01/22.
2. accumulation on 01/05/27.
3. value on 01/07/25.

Question 4 The effective rate of interest is 5.5% per time unit. You have a liability to pay the cashflows as shown in the following diagram (the time axis is calibrated in time units) at the end of the 1st, 2nd, 3rd and 4th time units:



In order to discharge this liability, you set up a bank account time $t = 0$ and pay into it an amount £X. You will meet the future payments out of this bank account.

Calculate the value of X that will allow you to exactly meet all the future payments.

Question 5

1. Describe in words the meaning of the following symbols:

   (a) $s^{i}_{\overline{10|}}$ (b) $\ddot{s}^{i}_{\overline{25|}}$ (c) $a^{i}_{\overline{10|}}$ (d) $\ddot{a}^{i}_{\overline{20|}}$

2. Assuming $i = 0.07$ per time unit, calculate the values of the above symbols.

Question 6 Prove the following:

1. $s_{\overline{n|}} = (1+i)^n a_{\overline{n|}}$.
2. $s_{\overline{n|}} = 1 + \ddot{s}_{\overline{n-1|}}$.
3. $\ddot{a}_{\overline{n|}} = v^n \ddot{s}_{\overline{n|}}$.
4. $\ddot{a}_{\overline{n|}} = 1 + a_{\overline{n-1|}}$.

(HINT: You should prove these from first principles)

Question 7 Calculate the value of the following annuity functions at the given interest rate. Give your answers to 5 decimal places.

1. $a^{0.075}_{\overline{11|}}$. 2. $\ddot{a}^{0.025}_{\overline{22|}}$. 3. $\ddot{s}^{0.09}_{\overline{8|}}$. 4. $s^{0.12}_{\overline{12|}}$.

Question 8 Assuming $i = 0.03$ per time unit,

1. Calculate the values of the perpetuities:

   (a) $a^{i}_{\overline{\infty|}}$. (b) $\ddot{a}^{i}_{\overline{\infty|}}$.

2. A payment stream starts at time $t = 0$. Annual payments of £1 are paid in arrears forever. Calculate the total value of the payment stream after 10 years (not only the remaining payments, but including all earlier payments too).

Question 9 Calculate the value of the following deferred annuities. Give your answers to 5 decimal places.

1. ${}_{10|}a^{0.045}_{\overline{10|}}$. 2. ${}_{5|}\ddot{a}^{0.08}_{\overline{14|}}$. 3. ${}_{9|}\ddot{s}^{0.025}_{\overline{17|}}$. 4. ${}_{20|}s^{0.09}_{\overline{25|}}$.

Question 10 Show that, if m and n are positive integers, then

$$a^{i}_{\overline{n+m|}} = a^{i}_{\overline{m|}} + {}_{m|}a^{i}_{\overline{n|}}.$$

Question 11 Calculate the value of the following annuity functions at the given interest rate. Give your answers to 5 decimal places..

1. $(Ia)^{0.10}_{\overline{5}|}$. 2. $(I\ddot{a})^{0.08}_{\overline{13}|}$. 3. $(I\ddot{s})^{0.005}_{\overline{34}|}$. 4. $(Is)^{0.05}_{\overline{15}|}$.

Question 12 Today, a person aged 60 has a pension payable yearly in arrears for 40 years. The first payment will be £15 000, and payments increase at 3% per annum. The rate of interest is 8% per annum effective.

1. Calculate the present value today of this annuity.
2. Calculate the accumulation at age 100 of this annuity.
3. Calculate the value at age 80 of this annuity.

Question 13 You have been awarded an annuity payable quarterly in advance. For the first 5 years the annual amount of the annuity will be £1 000, and for the next 10 years the annual amount of the annuity will be £1 500, after which payments will cease.

If the effective rate of interest is 3.5% per annum, calculate the present value now of this annuity.

Question 14 Suppose that time is measured in years. Starting at time 0, the effective rate of interest is 10% per annum for 10 years, then 12% per annum thereafter. Calculate the following:

1. The accumulation at the end of the payment stream of an annuity of £100 per annum, payable in arrears for 17 years.
2. The present value at the start of the payment stream of an annuity-due of £200 per month, payable for 20 years.
3. The present value at the start of the payment stream of a perpetuity of £1 000 per annum, payable yearly in advance.
4. The accumulation at the end of the payment stream of an increasing annuity payable yearly in arrears for 20 years, starting at £100 at the end of the first year and increasing by £100 each year.

Question 15 A man age exactly 20 today has been promised a pension by his employer, starting when he retires at age 65. It will be paid on each birthday from the 65th to the 100th inclusive.

The first payment will be 50% of his salary when he retires, and payments will then increase by 3% (compound) each year. His salary is currently £10 000 per annum, but it is expected that, due to inflation, this will increase by 2% each year (compound) until he retires at age 65.

The effective rate of interest will be 5% for the next 45 years and 6% thereafter. Calculate the present value today of this pension.

Answers: Q1.1. 184.00; Q1.2. 246.24; Q1.3. 137.50; Q2.1 -205.35; Q2.2 -139.75; Q2.3 -163.01; Q3.1 139.99; Q3.2 196.17; Q3.3 157.60; Q4.1 350.51; Q5.2 (a) 13.81645; Q5.2 (b) 67.67647; Q5.2 (c) 7.02358; Q5.2 (d) 11.33560; Q7.1. 7.31542; Q7.2. 17.18455; Q7.3. 12.02104; Q7.4. 24.13313; Q8.1.a. 33.33333; Q8.1.b. 34.33333; Q8.2. 44.79721; Q9.1. 5.09522; Q9.2. 7.36250; Q9.3. 21.38635; Q9.4. 84.70090; Q11.1. 10.65259; Q11.2. 50.70607; Q11.3. 632.22099; Q11.4 153.14984; Q12.1. £254 952.51; Q12.2. £5 538 721.28; Q12.3. £1 188 322.72; Q13. £15 345.91; Q14.1. £4 532.16; Q14.2. £21 096.39; Q14.3. £10 357.43; Q14.4. £48 984.88; Q15.1.£30 882.12.

F7AB - Actuarial and Financial Mathematics
Solutions to Tutorial 2 – Valuing cashflows and annuities

Solution 1

1. The accumulation to $t = 5$ is:

$$\begin{aligned} & 100V(5,0) - 20V(5,1) + 100V(5,2) - 20V(5,3) - 20V(5,4) \\ = \; & 100(1.06)^5 - 20(1.06)^4 + 100(1.06)^3 - 20(1.06)^2 - 20(1.06)^1 \\ = \; & £184.0026 \qquad (184.00 \text{ rounded}). \end{aligned}$$

2. The accumulation to $t = 10$ is the accumulation to $t = 5$ rolled up to time 10, i.e.

$$184.0026 \times V(10,5) = 184.0026 \times 1.06^5 = £246.2370 \qquad (246.24 \text{ rounded}).$$

3. The present value at $t = 0$ is the total accumulation to $t = 5$ rolled back to time 0, i.e.

$$184.0026 \times V(0,5) = 184.0026 \times 1.06^{-5} = 137.4974 \qquad (137.50 \text{ rounded}).$$

Solution 2

1. The accumulation to $t = 5$ is:

$$\begin{aligned} & 1000V(5,0) - 500V(5,1.5) + 100V(5,2) - 500V(5,2.5) - 500V(5,4) \\ = \; & 1000(1.08)^5 - 500(1.08)^{3.5} + 100(1.08)^3 - 500(1.08)^{2.5} - 500(1.08)^1 \\ = \; & -£205.3455 \qquad (-205.35 \text{ rounded}). \end{aligned}$$

2. The present value at $t = 0$ is the accumulation to $t = 5$ rolled back to time 0, i.e.

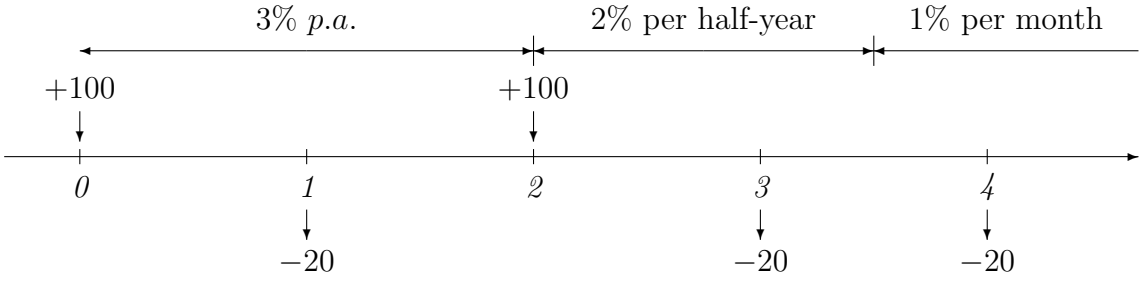
$$-205.3455 \times V(0,5) = -205.3455 \times 1.08^{-5} = -£139.7547 \qquad (-139.75 \text{ rounded}).$$

3. The value at $t = 2$ is the accumulation to $t = 5$ rolled back to time 2, i.e.

$$-205.3455 \times V(2,5) = -205.3455 \times 1.08^{-3} = -£163.0099 \qquad (-163.01 \text{ rounded})$$

Alternatively, show how can you obtain the value instead from the value at $t = 0$.

Solution 3 For convenience, we define the time unit to be 1 year, with 1/1/12 = 0, 1/1/13 = 1 and so on. We also add the interest rates to the diagram.



1. The present value on 01/01/12 ($t = 0$) is:

$$\begin{aligned}
&100V(0,0) - 20V(0,1) + 100V(0,2) - 200V(0,3) - 20V(0,4)\\
=&100 \times (1.03)^0 - 20 \times (1.03)^{-1} + 100 \times (1.03)^{-2} - 20 \times (1.03)^{-2} \times (1.02)^{-2}\\
&- 20 \times (1.03)^{-2} \times (1.02)^{-3} \times (1.01)^{-6}\\
=&£139.9872 \qquad (139.99 \text{ rounded}).
\end{aligned}$$

2. The accumulation on 01/05/17 ($t = 5\frac{4}{12}$) is the present value at $t = 0$ rolled up to time $5\frac{4}{12}$, i.e.

$$139.9872 \times V(5\frac{4}{12}, 0) = 139.9872 \times 1.03^2 \times 1.02^3 \times 1.01^{22} = £196.1704 \text{ (196.17 rounded)}.$$

The value on 01/07/15 ($t = 3\frac{6}{12}$) is the present value at $t = 0$ rolled up to time $3\frac{6}{12}$, i.e.

$$139.9872 \times V(3\frac{6}{12}, 0) = 139.9872 \times 1.03^2 \times 1.02^3 = £157.6026 \text{ (157.60 rounded)}.$$

Alternatively, show how can you obtain the value instead from the value at $t = 5\frac{4}{12}$.

Solution 4 If $£X$ is paid into the bank account at $t = 0$, and all the later payments are made from the bank account, the accumulation at $t = 4$ when the last withdrawal is made will be:

$$\begin{aligned}
& X\ V(4,0) - 100\ V(4,1) - 100\ V(4,2) - 100\ V(4,3) - 100\ V(4,4)\\
=\ & X\ (1.055)^4 - 100\ (1.055)^3 - 100\ (1.055)^2 - 100\ (1.055)^1 - 100\ (1.055)^0\\
=\ & X\ (1.055)^4 - 434.226637.
\end{aligned}$$

If the payments are met exactly, this accumulation will be zero. Therefore, $X = £350.5150$ (350.51 rounded).

Solution 5

1. (a) The accumulation to the end of the payment stream of an annuity of 1 unit per time unit, payable in arrears for 10 time units, at an effective rate of interest of i per time unit.

   (b) The accumulation to the end of the payment stream of an annuity of 1 unit per time unit, payable in advance for 25 time units, at an effective rate of interest of i per time unit.

   (c) The present value at the start of the payment stream of an annuity of 1 unit per time unit, payable in arrears for 10 time units, at an effective rate of interest of i per time unit.

   (d) The present value at the start of the payment stream of an annuity of 1 unit per time unit, payable in advance for 20 time units, at an effective rate of interest of i per time unit.

2. To 5 decimal places,

   (a) $\dfrac{1.07^{10} - 1}{0.07} = 13.81645.$

   (b) $1.07 \times \dfrac{1.07^{25} - 1}{0.07} = 67.67647.$

(c) $\dfrac{1-1.07^{-10}}{0.07} = 7.02358.$

(d) $1.07 \times \dfrac{1-1.07^{-20}}{0.07} = 11.33560.$

Solution 6

1. $$s_{\overline{n}|} = \sum_{r=1}^{n}(1+i)^{n-r} = (1+i)^n \sum_{r=1}^{n}(1+i)^{-r} = (1+i)^n a_{\overline{n}|}$$

2. $$s_{\overline{n}|} = \sum_{r=1}^{n}(1+i)^{n-r} = 1 + \sum_{r=1}^{n-1}(1+i)^{n-r} = 1 + \ddot{s}_{\overline{n-1}|}.$$

3. $$\ddot{a}_{\overline{n}|} = \sum_{r=0}^{n-1}(1+i)^{-r} = (1+i)^{-n} \sum_{r=1}^{n}(1+i)^{n-r-1} = v^n \ddot{s}_{\overline{n}|}$$

4. $$\ddot{a}_{\overline{n}|} = \sum_{r=0}^{n-1}(1+i)^{-r} = 1 + \sum_{r=1}^{n-1}(1+i)^{-r} = 1 + a_{\overline{n-1}|}.$$

Solution 7

1. $\dfrac{1-v^{11}}{i} = 7.31542.$
2. $\dfrac{1-v^{22}}{d} = 17.18455.$
3. $\dfrac{(1+i)^8-1}{d} = 12.02104.$
4. $\dfrac{(1+i)^{12}-1}{i} = 24.13313.$

Solution 8

1. (a) $\dfrac{1}{i} = 33.33333.$ (b) $\dfrac{1}{d} = 34.33333.$

2. $s^i_{\overline{10}|} + a^i_{\overline{\infty}|} = 11.46388 + 33.33333 = 44.79721.$

Solution 9

1. $v^{10}\, a^i_{\overline{10}|} = 5.09522.$
2. $v^5\, \ddot{a}^i_{\overline{14}|} = 7.36250.$
3. $\ddot{s}^i_{\overline{17}|} = 21.38635.$
4. $s^i_{\overline{25}|} = 84.70090.$

Solution 10

$$\begin{aligned} a^i_{\overline{n+m}|} &= v + v^2 + \cdots + v^m + v^{m+1} + \cdots + v^{m+n-1} + v^{m+n} \\ &= a^i_{\overline{m}|} + v^m(v + v^2 + \cdots + v^n) \\ &= a^i_{\overline{m}|} + {}_m|a^i_{\overline{n}|}. \end{aligned}$$

Solution 11

1. $\dfrac{\ddot{a}^i_{\overline{5}|} - 5v^5}{i} = 10.65259.$
2. $\dfrac{\ddot{a}^i_{\overline{13}|} - 13v^{13}}{d} = 50.70607.$
3. $\dfrac{\ddot{s}^i_{\overline{34}|} - 34}{d} = 632.22099.$
4. $\dfrac{\ddot{s}^i_{\overline{15}|} - 15}{i} = 153.14984.$

Solution 12

1. Set $r = 0.03$. To more easily calculate the present value (P.V.), we look for a geometric progression:

$$\begin{aligned} \text{P.V.} &= 15\,000 \times \left(v + (1+r)v^2 + (1+r)^2v^3 + \cdots + (1+r)^{39}v^{40}\right) \\ &= 15\,000 \times (1+r)^{-1}\left((1+r)v + (1+r)^2v^2 + (1+r)^3v^3 + \cdots + (1+r)^{40}v^{40}\right). \end{aligned}$$

Set $(1+j) = 1/(v(1+r)) = (1+i)/(1+r)$. Then $j = 0.048544$ (to 6 d.p.) and we can write

$$\begin{aligned} \text{P.V.} &= 15\,000 \times (1+r)^{-1}\left((1+j)^{-1} + (1+j)^{-2} + (1+j)^{-3} + \cdots + (1+j)^{-40}\right) \\ &= 15\,000 \times (1+r)^{-1} \times a^{j}_{\overline{40|}} \\ &= £254\,952.51. \end{aligned}$$

2. Accumulation at age 100 is the P.V. at age 60 multiplied by the factor $A(60, 100)$, i.e.

$$254\,952.51 \times (1+i)^{40} = 254\,952.51 \times 1.08^{40} = £5\,538\,721.28.$$

3. The value at age 80 is the P.V. at age 60 multiplied by the factor $A(60, 80)$, i.e.

$$254\,952.51 \times (1+i)^{20} = 254\,952.51 \times 1.08^{20} = £1\,188\,322.72.$$

Solution 13 The annuity is for £250 per quarter-year for 20 quarter-years, and then £375 per quarter-year for 40 quarter years. The effective interest rate is $i = 0.008637$ (6 d.p.) per quarter-year. Hence we can write the present value at $t = 0$ as:

$$250\,\ddot{a}^{0.008637}_{\overline{20|}} + 375\,{}_{20|}\ddot{a}^{0.008637}_{\overline{40|}} = 250 \times 18.45365 + 375 \times 1.008637^{-20} \times 33.99127 = £15\,345.91.$$

Solution 14

1. $100 \times (1.12^7\, s^{0.1}_{\overline{10|}} + s^{0.12}_{\overline{7|}}) = £4\,532.16.$
2. $200 \times (\ddot{a}^{0.007974}_{\overline{120|}} + 1.007974^{-120}\, \ddot{a}^{0.009489}_{\overline{120|}}) = £21\,096.39.$
3. $1,000 \times (\ddot{a}^{0.1}_{\overline{10|}} + 1.1^{-10}\, \ddot{a}^{0.12}_{\overline{\infty|}}) = £10\,357.43.$
4. $100 \times 1.12^{10}\, (Is)^{0.1}_{\overline{10|}} + 100 \times (Is)^{0.12}_{\overline{10|}} + 1\,000 \times s^{0.12}_{\overline{10|}} = £48\,984.88.$

Solution 15 The man retires in 45 years, when his salary is expected to be $£10\,000 \times 1.02^{45} = £24\,378.54$, so the first payment will be half of this, namely £12 189.27.

As the interest rate changes after age 65, we calculate the P.V. at age 65 first. Set $v = 1/1.06$. The present value at age 65 of his pension is

$$\text{P.V.} = 12\,189.27 \times \left(1 + 1.03v + 1.03^2v^2 + \cdots + 1.03^{35}v^{35}\right).$$

Note that there are 36 terms in the sum. Then setting $1 + j = 1.03v$ we can express the sum using the annuity notation, so that

$$\text{P.V.} = 12\,189.27 \times \ddot{a}^{j}_{\overline{36|}} = 12\,189.27 \times \ddot{a}^{0.029126}_{\overline{36|}} = £277\,476.08.$$

Finally, the present value of the annuity at age 20 is $277\,476.08 \times 1.05^{-45} = £30\,882.12$.